



Microcontrollers

Rex Joseph

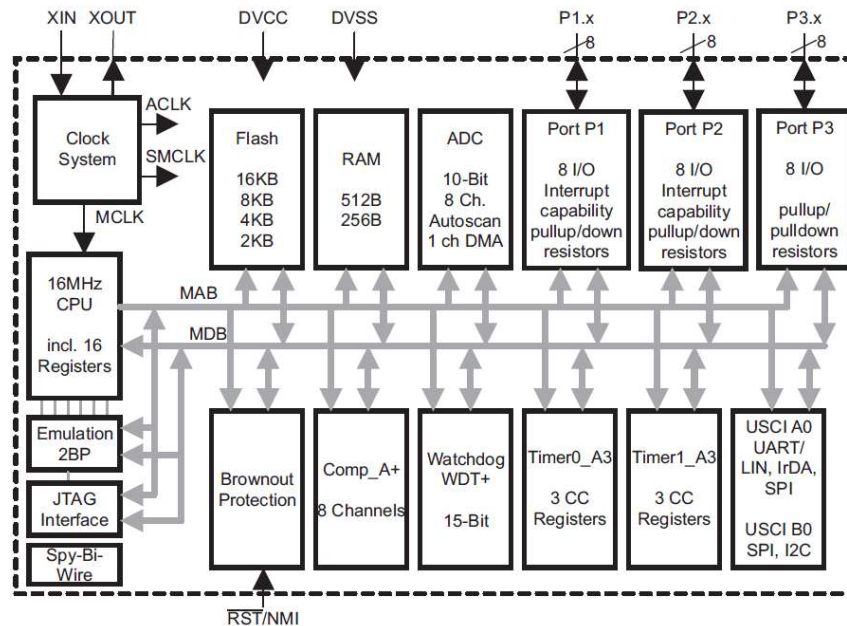
Department of Electrical and
Electronics Engineering

MSP430 Architecture - Recap



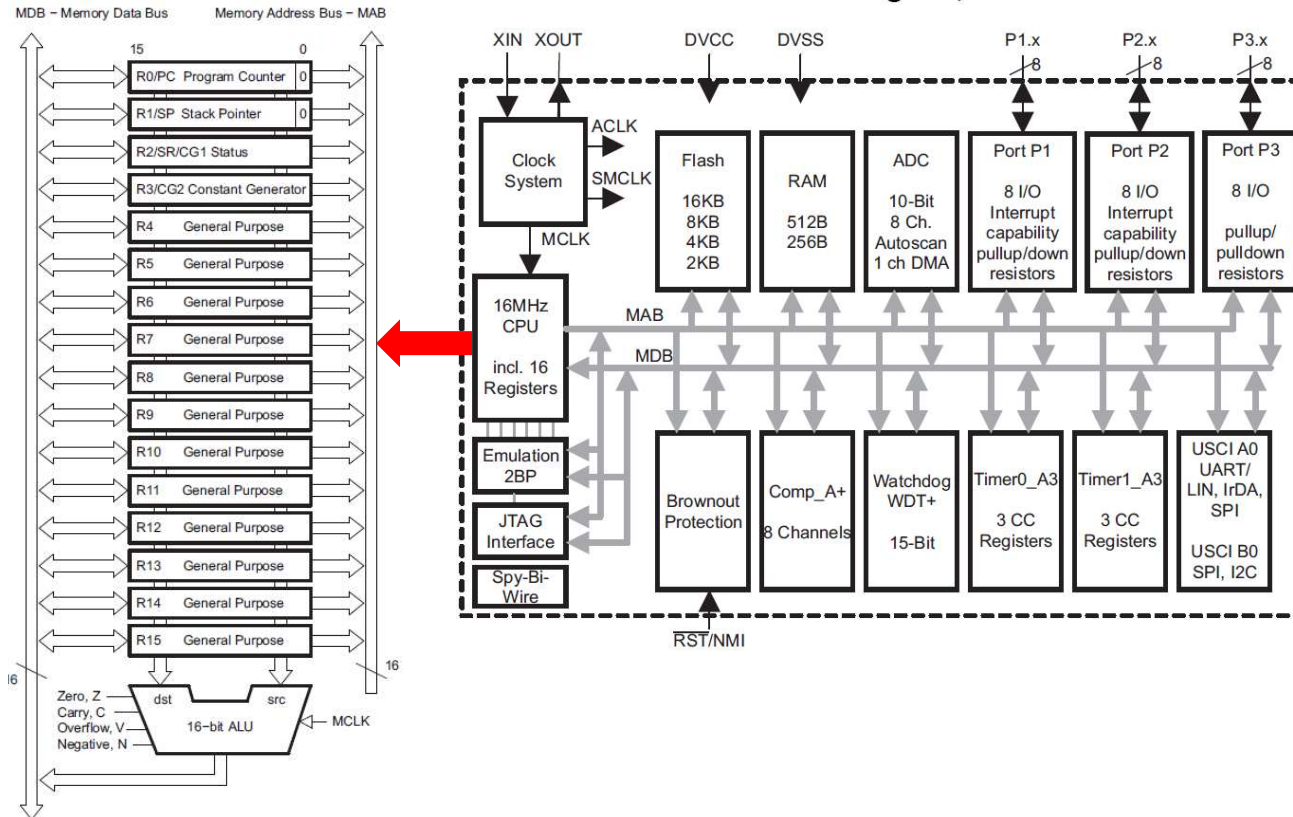
MSP430 Architecture - Recap

Functional Block Diagram, MSP430G2x53



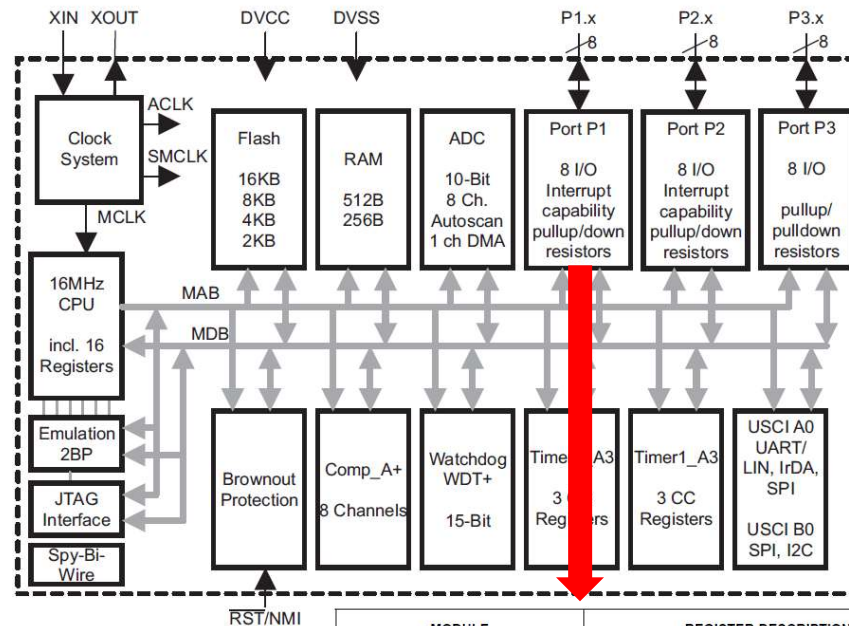
MSP430 Architecture - CPU

Functional Block Diagram, MSP430G2x53



MSP430 Architecture – Port 1

Functional Block Diagram, MSP430G2x53

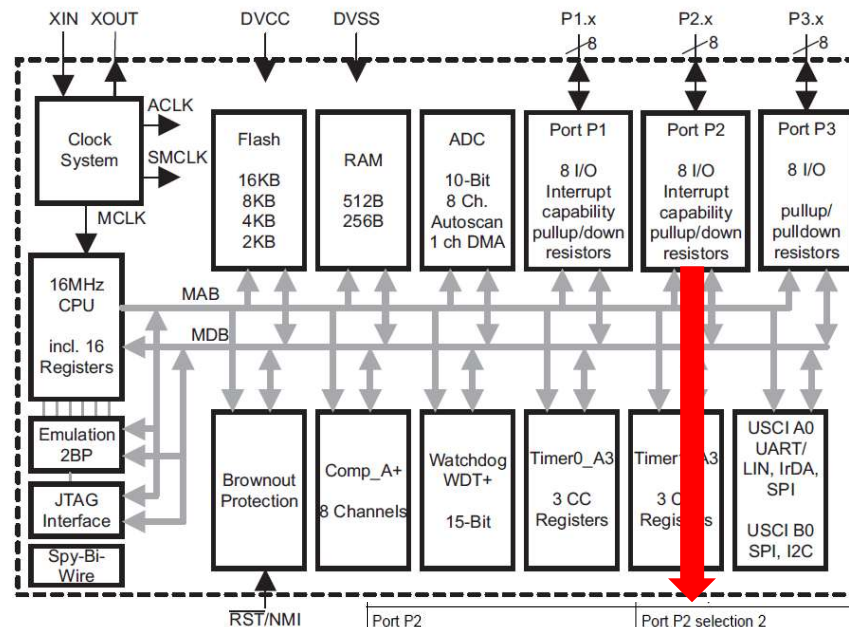


8 bit Registers (SFR) for port 1
9 in all

MODULE	REGISTER DESCRIPTION	REGISTER NAME	OFFSET
Port P1	Port P1 selection 2	P1SEL2	041h
	Port P1 resistor enable	P1REN	027h
	Port P1 selection	P1SEL	026h
	Port P1 interrupt enable	P1IE	025h
	Port P1 interrupt edge select	P1IES	024h
	Port P1 interrupt flag	P1IFG	023h
	Port P1 direction	P1DIR	022h
	Port P1 output	P1OUT	021h
	Port P1 input	P1IN	020h

MSP430 Architecture – Port 2

Functional Block Diagram, MSP430G2x53

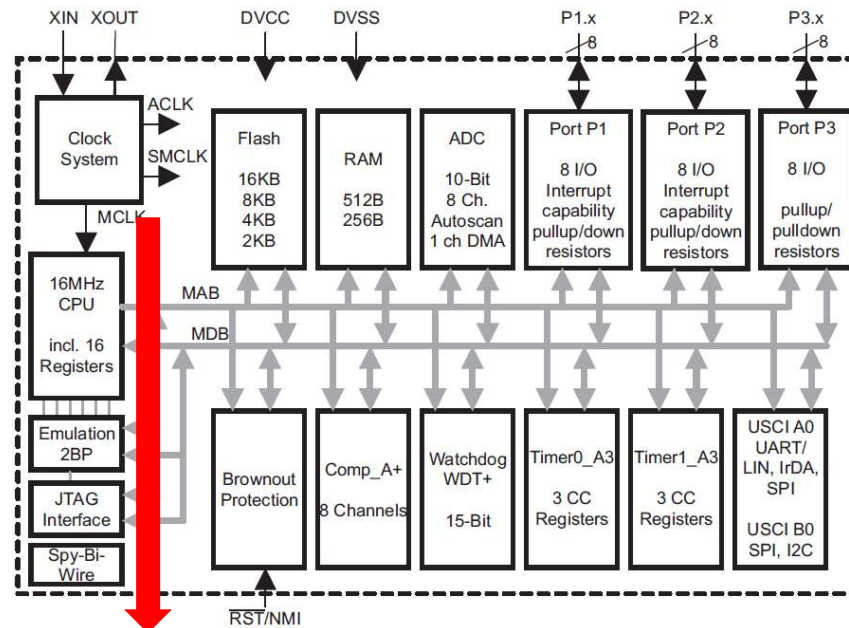


8 bit Registers (SFR) for port 2
9 in all

Port P2	Port P2 selection 2	P2SEL2	042h
	Port P2 resistor enable	P2REN	02Fh
	Port P2 selection	P2SEL	02Eh
	Port P2 interrupt enable	P2IE	02Dh
	Port P2 interrupt edge select	P2IES	02Ch
	Port P2 interrupt flag	P2IFG	02Bh
	Port P2 direction	P2DIR	02Ah
	Port P2 output	P2OUT	029h
	Port P2 input	P2IN	028h

MSP430 Architecture - CPU

Functional Block Diagram, MSP430G2x53

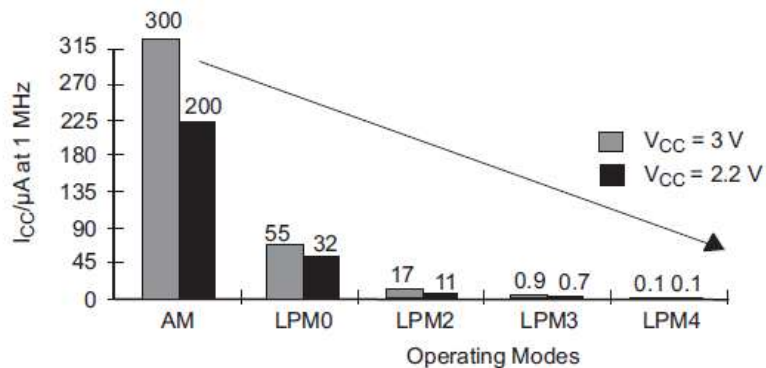


Basic Clock System+	Basic clock system control 3	BCSCTL3	053h
	Basic clock system control 2	BCSCTL2	058h
	Basic clock system control 1	BCSCTL1	057h
	DCO clock frequency control	DCOCTL	056h

MSP430 Architecture – Operating modes

Table 2-2. Operating Modes For Basic Clock System

SCG1	SCG0	OSCOFF	CPUOFF	Mode	CPU and Clocks Status
0	0	0	0	Active	CPU is active, all enabled clocks are active
0	0	0	1	LPM0	CPU, MCLK are disabled, SMCLK, ACLK are active
0	1	0	1	LPM1	CPU, MCLK are disabled. DCO and DC generator are disabled if the DCO is not used for SMCLK. ACLK is active.
1	0	0	1	LPM2	CPU, MCLK, SMCLK, DCO are disabled. DC generator remains enabled. ACLK is active.
1	1	0	1	LPM3	CPU, MCLK, SMCLK, DCO are disabled. DC generator disabled. ACLK is active.
1	1	1	1	LPM4	CPU and all clocks disabled



MSP430 Architecture – Operating modes

The low-power modes 0 to 4 are configured with the CPUOFF, OSCOFF, SCG0, and SCG1 bits in the status register

The advantage of including the CPUOFF, OSCOFF, SCG0, and SCG1 mode-control bits in the status register is that the present operating mode is saved onto the stack during an interrupt service routine.

Program flow returns to the previous operating mode if the saved SR value is not altered during the interrupt service routine.

Program flow can be returned to a different operating mode by manipulating the saved SR value on the stack inside of the interrupt service routine.

The mode-control bits and the stack can be accessed with any instruction.

MSP430 Architecture – Operating modes

When setting any of the mode-control bits, the selected operating mode takes effect immediately

Peripherals operating with any disabled clock are disabled until the clock becomes active.

The peripherals may also be disabled with their individual control register settings.

MSP430 Architecture – Operating modes

Entering and Exiting Low-Power Modes

An enabled interrupt event wakes the MSP430 from any of the low-power operating modes. The program flow is:

- Enter interrupt service routine:
 - The PC and SR are stored on the stack
 - The CPUOFF, SCG1, and OSCOFF bits are automatically reset

- Options for returning from the interrupt service routine:
 - The original SR is popped from the stack, restoring the previous operating mode.
 - The SR bits stored on the stack can be modified within the interrupt service routine returning to a different operating mode when the RETI instruction is executed.

MSP430 Architecture – Operating modes

; Enter LPM0 Example

BIS #GIE+CPUOFF,SR

; ...

; Exit LPM0 Interrupt Service Routine

BIC #CPUOFF,0(SP)

RETI

; Enter LPM3 Example

BIS #GIE+CPUOFF+SCG1+SCG0,SR

; ...

;

; Exit LPM3 Interrupt Service Routine

BIC #CPUOFF+SCG1+SCG0,0(SP) ; Exit LPM3 on RETI

RETI

; Enter LPM0

; Program stops here

;

; Exit LPM0 on RETI

; Enter LPM3

; Program stops here

MSP430 Architecture – Clock Module

The basic clock module+ supports low system cost and ultra-low power consumption.

Three internal clock signals to select the best balance of performance and low power consumption.

The basic clock module can be configured to operate without any external components, with one external resistor, with one or two external crystals, or with resonators, **under full software control**.

MSP430 Architecture – Clock Module

The basic clock module includes two, three or four clock sources:

- **LFXT1CLK**: Low-frequency/high-frequency oscillator that can be used with low-frequency watch crystals or external clock sources of 32768 Hz *or with standard crystals, resonators, or external clock sources in the 400-kHz to 16-MHz range.*
- **XT2CLK**: Optional high-frequency oscillator that can be used with standard crystals, resonators, or external clock sources in the 400-kHz to 16-MHz range.
- **DCOCLK**: Internal digitally controlled oscillator (DCO).
- **VLOCLK**: Internal very low-power low-frequency oscillator with 12-kHz typical frequency.

MSP430F20xx, MSP430G2xx1, MSP430G2xx2, MSP430G2xx3: LFXT1 does not support HF mode, XT2 is not present, ROSC is not supported.

MSP430 Architecture – Clock Module

Three clock signals are available from the basic clock module:

- **ACLK:** Auxiliary clock. ACLK is software selectable as LFXT1CLK or VLOCLK. ACLK is divided by 1, 2, 4, or 8.

ACLK is software selectable for individual peripheral modules.

- **MCLK:** Master clock. MCLK is software selectable as LFXT1CLK, VLOCLK, XT2CLK (if available on-chip), or DCOCLK.

MCLK is divided by 1, 2, 4, or 8.

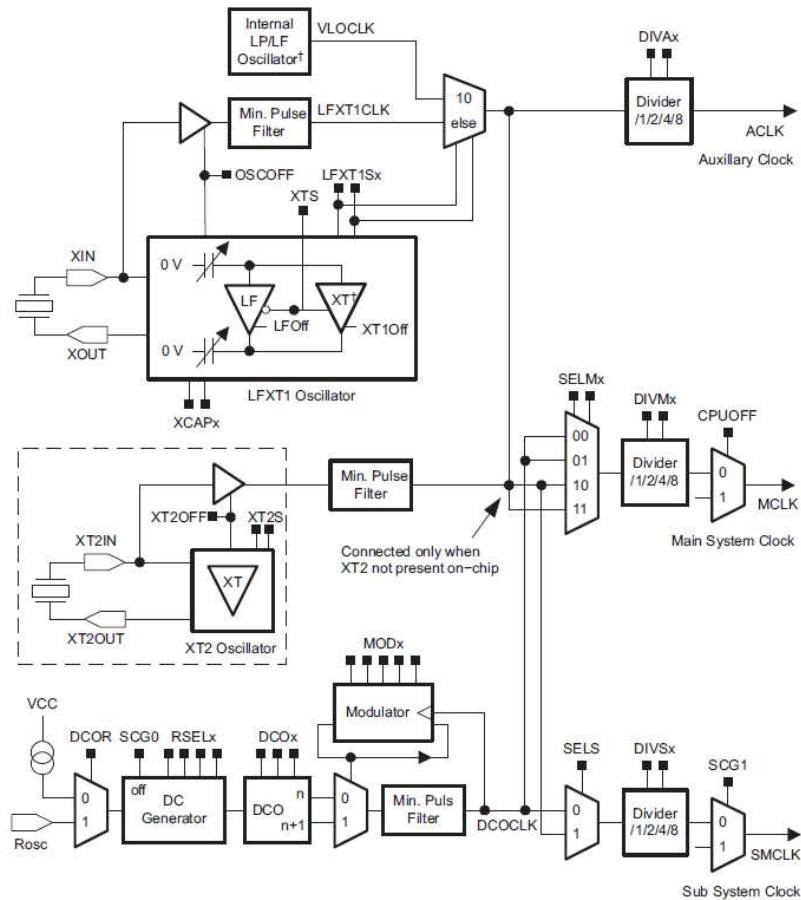
MCLK is used by the CPU and system.

- **SMCLK:** Sub-main clock. SMCLK is software selectable as LFXT1CLK, VLOCLK, XT2CLK (if available onchip), or DCOCLK.

SMCLK is divided by 1, 2, 4, or 8.

SMCLK is software selectable for individual peripheral modules.

MSP430 Architecture – Clock Module



MSP430 Architecture – Clock Module

Basic Clock Module Operation

After a PUC, **MCLK** and **SMCLK** are sourced from DCOCLK at ~**1.1 MHz** (device-specific data sheet for parameters) and **ACLK** is sourced from LFXT1CLK in LF mode with an internal load capacitance of 6 pF.

Status register control bits SCG0, SCG1, OSCOFF, and CPUOFF configure the MSP430 operating modes and enable or disable portions of the basic clock module

The **DCOCTL**, **BCSCTL1**, **BCSCTL2**, and **BCSCTL3** registers configure the basic clock module.

MSP430 Architecture – Clock Module

The basic clock module can be configured or reconfigured by software at any time during program execution, for example:

<i>CLR.B &DCOCTL</i>	<i>; Select lowest DCOx</i>
	<i>; and MODx settings</i>
<i>BIS.B #RSEL2+RSEL1+RSEL0,&BCSCTL1</i>	<i>; Select range 7</i>
<i>BIS.B #DCO2+DCO1+DCO0,&DCOCTL</i>	<i>; Select max DCO tap</i>

Conflicting requirements typically exist in battery-powered applications:

- Low clock frequency for energy conservation and time keeping
- High clock frequency for fast reaction to events and fast burst processing capability
- Clock stability over operating temperature and supply voltage

MSP430 Architecture – Clock Module

The internal **very-low-power low-frequency oscillator** (VLO) provides a typical frequency of 12 kHz (device specific parameters) without requiring a crystal

The **LFXT1 oscillator** supports ultra-low current consumption using a 32768-Hz watch crystal in LF mode (XTS = 0). LFXT1 may be used with an **external clock signal** on the XIN pin in either LF or HF mode

A watch crystal connects to XIN and XOUT without any other external components.

The software selectable XCAPx bits configure the internally provided load capacitance for the LFXT1 crystal in LF mode.

This capacitance can be selected as 1 pF, 6 pF, 10 pF, or 12.5 pF typical. Additional external capacitors can be added if necessary.

MSP430 Architecture – Clock Module

Digitally Controlled Oscillator (DCO)

The DCO is an integrated digitally controlled oscillator.

The DCO frequency can be adjusted by software using the **DCOx**, **MODx**, and **RSELx** bits

Adjusting the DCO Frequency

After a PUC, **RSELx** = 7 and **DCOx** = 3, allowing the DCO to start at a mid-range frequency. **MCLK** and **SMCLK** are sourced from **DCOCLK**.

Because the **CPU executes code from MCLK**, which is sourced from the fast-starting DCO, code execution typically begins from PUC in less than **2 μ s**.

MSP430 Architecture – Clock Module

The frequency of DCOCLK is set by the following functions:

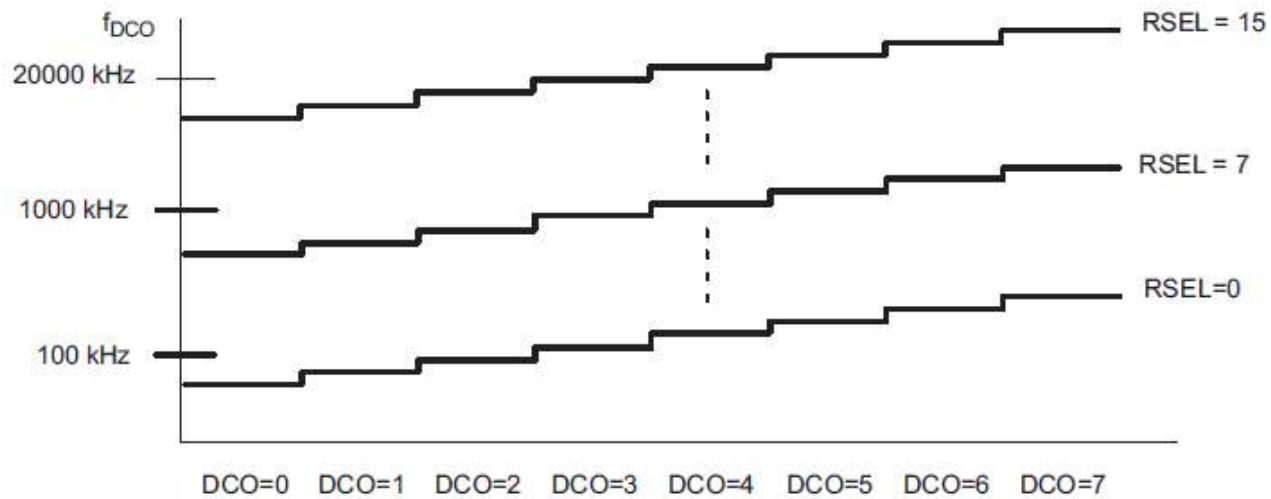
- The **four RSELx bits** select one of sixteen nominal frequency ranges for the DCO.

These ranges are defined for an individual device in the device-specific data sheet.

- The **three DCOx** bits divide the DCO range selected by the RSELx bits into 8 frequency steps, separated by approximately 10%.
- The **five MODx** bits, switch (**modulate**) between the frequency selected by the DCOx bits **and the next higher frequency** set by DCOx+1.

When DCOx = 07h, the MODx bits have no effect because the DCO is already at the highest setting for the selected RSELx range.

MSP430 Architecture – Clock Module



MSP430 Architecture – Clock Module

Each MSP430F2xx device (and most MSP430G2xx devices) has calibrated DCOCTL and BCSCTL1 register settings for specific frequencies stored in **information memory segment A**.

To use the calibrated settings, the information is copied into the DCOCTL and BCSCTL1 registers.

The calibrated settings affect the DCOx, MODx, and RSELx bits, and clear all other bits, except XT2OFF which remains set.

The remaining bits of BCSCTL1 can be set or cleared as needed with BIS.B or BIC.B instructions.

; Set DCO to 1 MHz:

CLR.B &DCOCTL

; and MODx settings

MOV.B &CALBC1_1MHZ,&BCSCTL1

MOV.B &CALDCO_1MHZ,&DCOCTL

; Select lowest DCOx

; Set range

; Set DCO step + modulation

MSP430 Architecture – Clock Module

DCO Modulator

The modulator mixes two DCO frequencies, f_{DCO} and $f_{\text{DCO}+1}$ to produce an intermediate effective frequency between f_{DCO} and $f_{\text{DCO}+1}$ and spread the clock energy, reducing electromagnetic interference (EMI).

The modulator mixes f_{DCO} and $f_{\text{DCO}+1}$ for 32 DCOCLK clock cycles and is configured with the MODx bits.

When MODx = 0 the modulator is off.

The modulator mixing formula is:

$$t = (32 - \text{MODx}) \times t_{\text{DCO}} + \text{MODx} \times t_{\text{DCO}+1}$$

Because f_{DCO} is lower than the effective frequency and $f_{\text{DCO}+1}$ is higher than the effective frequency, the error of the effective frequency integrates to zero.

It does not accumulate

MSP430 Architecture – Clock Module

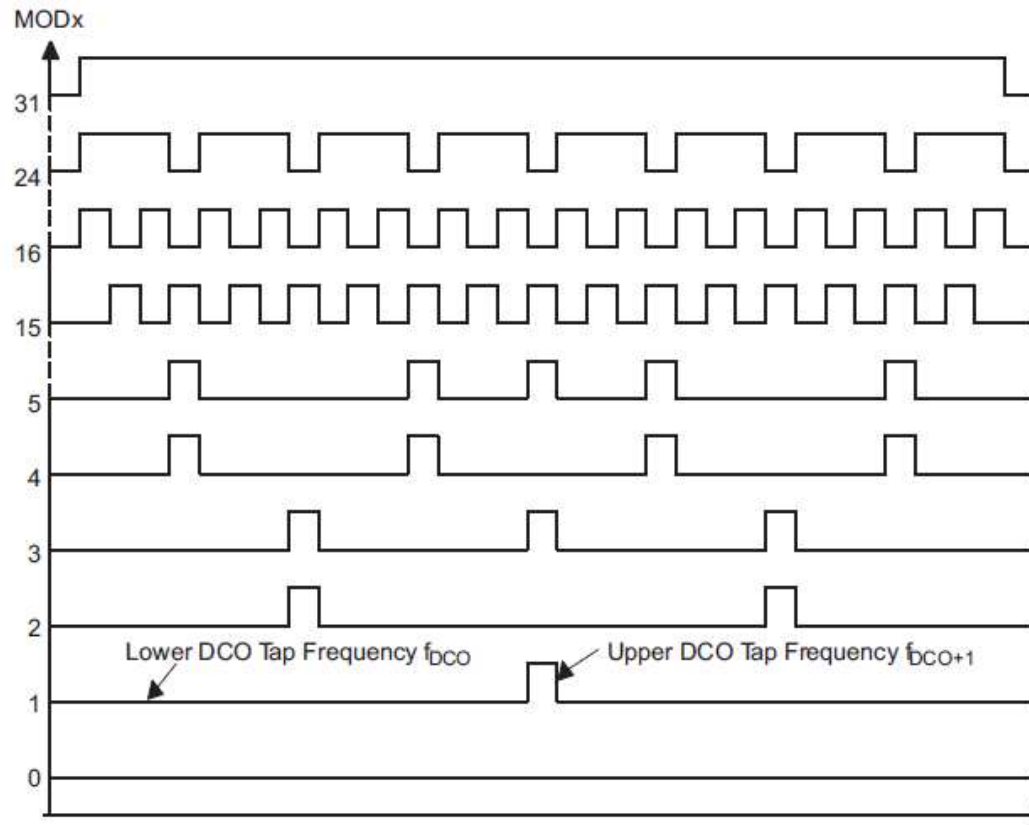


Figure 5-7. Modulator Patterns

MSP430 Architecture – Clock Module

Basic Clock Module Fail-Safe Operation

The basic clock module incorporates an oscillator-fault fail-safe feature. This feature detects an oscillator fault for LFXT1 and XT2 as shown in

The available fault conditions are:

- Low-frequency oscillator fault (LFXT1OF) for LFXT1 in LF mode
- High-frequency oscillator fault (LFXT1OF) for LFXT1 in HF mode
- High-frequency oscillator fault (XT2OF) for XT2

The crystal oscillator fault bits LFXT1OF, and XT2OF are set if the corresponding crystal oscillator is turned on and not operating properly.

The fault bits remain set as long as the fault condition exists and are automatically cleared if the enabled oscillators function normally.

MSP430 Architecture – Clock Module

The OFIFG oscillator fault flag is set and latched at POR or when an oscillator fault (LFXT1OF, or XT2OF) is detected.

When OFIFG is set, MCLK is sourced from the DCO, and if OFIE is set, the OFIFG requests an NMI interrupt.

When the interrupt is granted, the OFIE is reset automatically.
The OFIFG flag must be cleared by software.

The source of the fault can be identified by checking the individual fault bits.

If a fault is detected for the crystal oscillator sourcing the MCLK, the MCLK is automatically switched to the DCO for its clock source.

This does not change the SELMx bit settings.

This condition must be handled by user software.

MSP430 Architecture – Clock Module

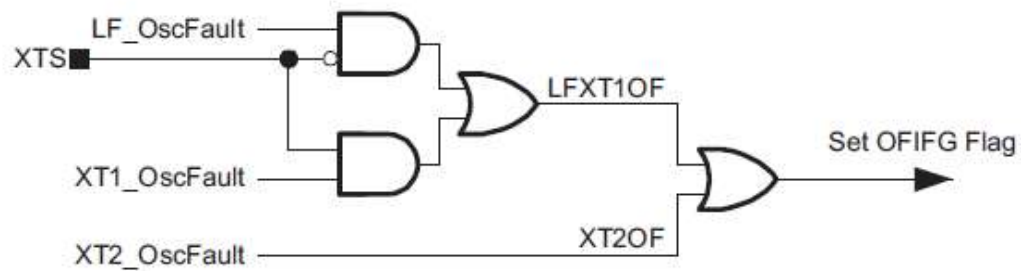


Figure 5-8. Oscillator-Fault Logic

MSP430 Architecture – Clock Module

Basic Clock Module+ Registers

Table below lists the memory-mapped registers for the Basic Clock Module+.

Table 5-1. Basic Clock Module+ Registers

Address	Acronym	Register Name	Type	Reset
56h	DCOCTL	DCO control	Read/write	60h with PUC
57h	BCSCTL1	Basic clock system control 1	Read/write	87h with POR ⁽¹⁾
58h	BCSCTL2	Basic clock system control 2	Read/write	00h with PUC
53h	BCSCTL3	Basic clock system control 3	Read/write	05h with PUC ⁽²⁾
00h	IE1	SFR interrupt enable 1	Read/write	00h with PUC
02h	IFG1	SFR interrupt flag 1	Read/write	02h with PUC

MSP430 Architecture – Clock Module

Figure 5-10. DCOCTL Register

7	6	5	4	3	2	1	0
DCOx			MODx				
rw-0	rw-1	rw-1	rw-0	rw-0	rw-0	rw-0	rw-0

Table 5-2. DCOCTL Register Field Descriptions

Bit	Field	Type	Reset	Description
7-5	DCOx	R/W	3h	DCO frequency select. These bits select which of the eight discrete DCO frequencies within the range defined by the RSELx setting is selected.
4-0	MODx	R/W	0h	Modulator selection. These bits define how often the f_{DCO+1} frequency is used within a period of 32 DCOCLK cycles. During the remaining clock cycles (32 – MOD) the f_{DCO} frequency is used. Not useable when DCOx = 7.

MSP430 Architecture – Clock Module

Figure 5-11. BCSCTL1 Register

7	6	5	4	3	2	1	0
XT2OFF	XTS	DIVAx		RSELx			
rw-(1)	rw-(0)	rw-(0)	rw-(0)	rw-0	rw-1	rw-1	rw-1

Table 5-3. BCSCTL1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7	XT2OFF	R/W	1h	XT2 off. This bit turns off the XT2 oscillator. 0b = XT2 is on 1b = XT2 is off if it is not used for MCLK or SMCLK.
6	XTS ^{(1) (2)}	R/W	0h	LFXT1 mode select 0b = Low-frequency mode 1b = High-frequency mode
5-4	DIVAx	R/W	0h	Divider for ACLK 00b = /1 01b = /2 10b = /4 11b = /8
3-0	RSELx	R/W	7h	Range select. Sixteen different frequency ranges are available. The lowest frequency range is selected by setting RSELx = 0. RSEL3 is ignored when DCOR = 1.

MSP430 Architecture – Clock Module

Figure 5-12. BCSTL2 Register

7	6	5	4	3	2	1	0
SELMx		DIVMx		SELS	DIVSx		DCOR
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 5-4. BCSTL2 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	SELMx	R/W	0h	Select MCLK. These bits select the MCLK source. 00b = DCOCLK 01b = DCOCLK 10b = XT2CLK when XT2 oscillator present on-chip. LFXT1CLK or VLOCLK when XT2 oscillator not present on-chip. 11b = LFXT1CLK or VLOCLK
5-4	DIVMx	R/W	0h	Divider for MCLK 00b = /1 01b = /2 10b = /4 11b = /8
3	SELS	R/W	0h	Select SMCLK. This bit selects the SMCLK source. 0b = DCOCLK 1b = XT2CLK when XT2 oscillator present. LFXT1CLK or VLOCLK when XT2 oscillator not present
2-1	DIVSx	R/W	0h	Divider for SMCLK 00b = /1 01b = /2 10b = /4 11b = /8
0	DCOR ^{(1) (2)}	R/W	0h	DCO resistor select. Not available in all devices. See the device-specific data sheet. 0b = Internal resistor 1b = External resistor

MSP430 Architecture – Clock Module

Figure 5-13. BCSCTL3 Register

7	6	5	4	3	2	1	0
XT2Sx		LFXT1Sx		XCAPx		XT2OF	LFXT1OF
rw-0	rw-0	rw-0	rw-0	rw-0	rw-1	r-0	r-(1)

Table 5-5. BCSCTL3 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-6	XT2Sx	R/W	0h	XT2 range select. These bits select the frequency range for XT2. 00b = 0.4- to 1-MHz crystal or resonator 01b = 1- to 3-MHz crystal or resonator 10b = 3- to 16-MHz crystal or resonator 11b = Digital external 0.4- to 16-MHz clock source
5-4	LFXT1Sx ⁽¹⁾	R/W	0h	Low-frequency clock select and LFXT1 range select. These bits select between LFXT1 and VLO when XTS = 0, and select the frequency range for LFXT1 when XTS = 1. MSP430G22x0: The LFXT1Sx bits should be programmed to 10b during the initialization and start-up code to select VLOCLK (for more details refer to Digital I/O chapter). The other bits are reserved and should not be altered. When XTS = 0: 00b = 32768-Hz crystal on LFXT1 01b = Reserved 10b = VLOCLK (Reserved in MSP430F21x1 devices) 11b = Digital external clock source When XTS = 1 (Not applicable for MSP430F20xx, MSP430G2xx1, MSP430G2xx2, MSP430G2xx3): 00b = 0.4- to 1-MHz crystal or resonator 01b = 1- to 3-MHz crystal or resonator 10b = 3- to 16-MHz crystal or resonator 11b = Digital external 0.4- to 16-MHz clock source LFXT1Sx definition for MSP430AFE2xx devices: 00b = Reserved 01b = Reserved 10b = VLOCLK 11b = Reserved

MSP430 Architecture – Clock Module

Table 5-5. BCSCTL3 Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3-2	XCAPx ⁽²⁾	R/W	1h	Oscillator capacitor selection. These bits select the effective capacitance seen by the LFXT1 crystal when XTS = 0. If XTS = 1 or if LFXT1Sx = 11 XCAPx should be 00. This bit is reserved in the MSP430AFE2xx devices. 00b = Approximately 1 pF 01b = Approximately 6 pF 10b = Approximately 10 pF 11b = Approximately 12.5 pF
1	XT2OF ⁽³⁾	R	0h	XT2 oscillator fault. Does not apply to MSP430x2xx, MSP430x21xx, or MSP430x22xx devices. 0b = No fault condition present 1b = Fault condition present
0	LFXT1OF ⁽²⁾	R	1h	LFXT1 oscillator fault. This bit is reserved in the MSP430AFE2xx devices. 0b = No fault condition present 1b = Fault condition present

(1) MSP430G22x0: The LFXT1Sx bits should be programmed to 10b during the initialization and start-up code to select VLOCLK (for more details, refer to the Digital I/O chapter). The other bits are reserved and should not be altered.

(2) This bit is reserved in the MSP430AFE2xx devices.

(3) Does not apply to MSP430x2xx, MSP430x21xx, or MSP430x22xx devices.

MSP430 Architecture – Clock Module

Figure 5-14. IE1 Register

7	6	5	4	3	2	1	0
						OFIE	
rw-0							

Table 5-6. IE1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-2				These bits may be used by other modules. See device-specific data sheet.
1	OFIE ⁽¹⁾	R/W	0h	Oscillator fault interrupt enable. This bit enables the OFIFG interrupt. Because other bits in IE1 may be used for other modules, it is recommended to set or clear this bit using BIS.B or BIC.B instructions, rather than MOV.B or CLR.B instructions. MSP430G22x0: This bit should not be set. 0b = Interrupt not enabled 1b = Interrupt enabled
0				This bit may be used by other modules. See device-specific data sheet.

MSP430 Architecture – Clock Module

Figure 5-15. IFG1 Register

7	6	5	4	3	2	1	0
						OFIFG	
rw-1							

Table 5-7. IFG1 Register Field Descriptions

Bit	Field	Type	Reset	Description
7-2				These bits may be used by other modules. See device-specific data sheet.
1	OFIFG ⁽¹⁾	R/W	1h	Oscillator fault interrupt flag. Because other bits in IFG1 may be used for other modules, it is recommended to set or clear this bit using BIS.B or BIC.B instructions, rather than MOV.B or CLR.B instructions. 0b = Interrupt not pending 1b = Interrupt pending
0				This bit may be used by other modules. See device-specific data sheet.



THANK YOU

Rex Joseph

Department of Electrical and Electronics Engineering

rexjoseph@pes.edu

+91 80 6666 3333 Extn 515